

Proposal: Expanding the Heatmap Value Metrics for Market-Regime Research

Objective

Expand the Heatmap **Value** dropdown beyond static butterfly pricing so the options surface can be studied as a real-time indicator of market state.

The central research question is:

What can the butterfly surface tell us about the actual intraday market regime that VIX, spot, or a static gamma measurement cannot?

Proposed Value Metrics

Metric	What It Measures	Research Purpose
Debit	Current butterfly price	Raw surface
Credit	Current liquidation/exit value	Execution reality
% Change	Change from previous observation	Surface movement
R:R	Reward relative to capital at risk	Opportunity quality
Δ Debit	Dollar change over a fixed interval	First derivative — where the surface is moving
Δ^2 Debit	Change in Δ Debit	Second derivative — acceleration/ deceleration
Velocity	Debit change per minute	Speed of repricing
Acceleration	Change in velocity	Potential regime-transition signal
Slope	Value change between adjacent center strikes	Surface gradient
Curvature	Change in slope across strikes	Shape and bending of the surface
Call/Put Asymmetry	Equivalent call vs. put fly values	Directional/skew imbalance
Width Efficiency	Value response normalized by fly width	Comparison across 20/25/30/35/40/etc. widths
Time Decay	Value change attributable to time	Separates theta from market movement

Metric	What It Measures	Research Purpose
Spot Sensitivity	Fly-value change per SPX point	Responsiveness of the surface to spot
Surface Stability	Persistence of surface shape through time	Stable regime vs. reorganizing regime

Core Research Stack

The highest-priority metrics would be:

1. **Debit** — What does the surface look like?
2. **Δ Debit** — Where is it moving?
3. **Δ^2 Debit** — Is that movement accelerating?
4. **Curvature** — What shape/regime is developing?
5. **Call/Put Asymmetry** — Is repricing symmetric or directional?

Together, these transform the heatmap from a static butterfly-price display into a potential **market-state sensor**.

Gamma Relationship

Gamma/GEX describes **potential dealer hedging pressure**. It does not represent all market flow, and dealers are not always the dominant force.

The butterfly surface provides another perspective: **what the options market is actually pricing and how that pricing is changing in real time**.

The research should therefore test whether changes in **surface velocity, acceleration, curvature, and asymmetry** precede observable changes in SPX behavior—and whether those changes help identify when the apparent gamma regime is actually controlling price versus when outside flows are overwhelming it.

Potential Composite Metric: Surface Regime Score

Ultimately, the individual measurements could potentially be combined into a **Surface Regime Score (SRS)** using:

Velocity + Acceleration + Curvature + Asymmetry + Stability

The objective would be to classify the current surface into states such as:

Stable/Compressed → Expanding → Directional → Unstable/Transitioning

This would provide a testable framework for determining whether the butterfly surface itself can identify intraday market-regime changes.